MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE BMB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.1209 failures per million operating hours for class I or 0.6813 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BMB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/254-1 for class I and M28787/254-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3 to 18 Vdc ±5%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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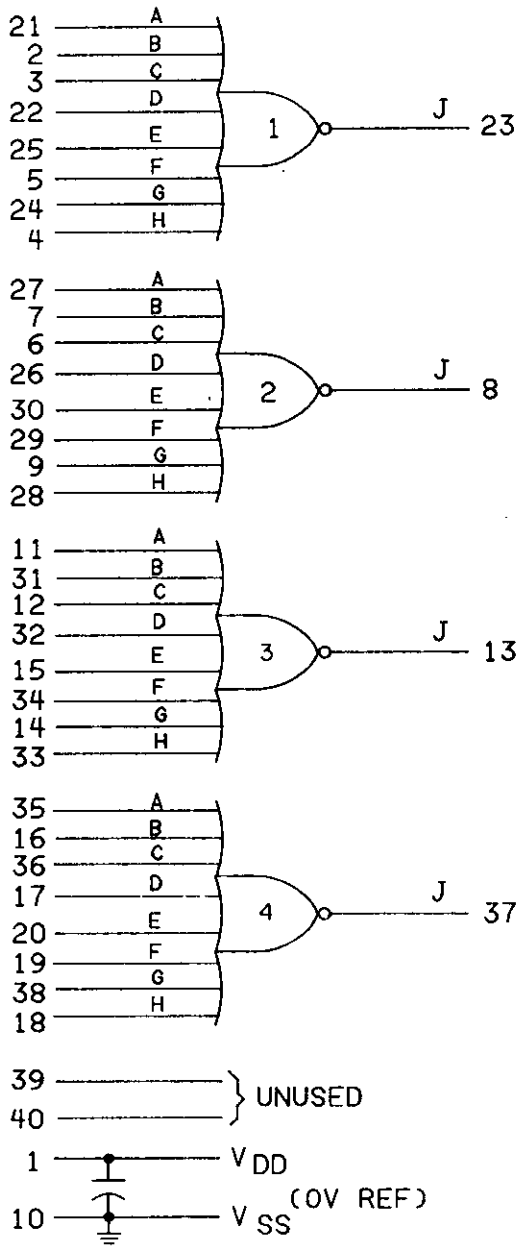


FIGURE 1. BMB Module function diagram, NAND gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ -3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation Initial End-of-life	III III	S S	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	MΩ MΩ
B1-B32	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C32	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D4	Static output voltage, high level (I_{OH}= 0)	VI	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
E1-E4	Static output voltage, low level (I_{OL}= 0)	VII	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F36	Static output voltage, high level	VIII	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
F37-F40	Static output voltage, low level	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
G1-G4	Static output voltage, high level (V_{DD} = 15V)	IX	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
G5-G8	Static output voltage, low level (V_{DD} = 15V)	IX	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
H1-H32	Input current, high level	X	X	0	200	---	---	0	200	---	---	0	200	nA
J1-J32	Input current, low level	XI	X	0	-200	---	---	0	-200	---	---	0	-200	nA
K1	Power supply current, low level	XII	X	0	8	0	8	0	75	0	8	0	160	µA
K2	Power supply current, high level	XII	X	0	8	0	8	0	75	0	8	0	160	µA
L1-L4	Transition time (t_{TLH}), C_L= 50 pF, V_{DD}= 5V	XIII	S	20	200	20	200	22	220	15	152	25	245	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ -3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L5-L8	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5V	XIII	Q	65	680	65	680	75	750	50	517	80	835	ns
M1-M4	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5V	XIV	S	20	200	20	200	22	220	15	152	25	245	ns
M5-M8	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5V	XIV	Q	65	680	65	680	75	750	50	517	80	835	ns
N1-N4	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10V	XV	Q	10	100	10	100	11	110	7	75	12	125	ns
N5-N8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10V	XV	Q	10	100	10	100	11	110	7	75	12	125	ns
P1-P4	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15V	XVI	Q	8	80	8	80	9	90	6	60	10	100	ns
P5-P8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15V	XVI	Q	8	80	8	80	9	90	6	60	10	100	ns
Q1-Q32	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	XVII	S	50	500	50	500	55	555	38	380	61	615	ns
Q33-Q64	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	XVII	Q	70	700	70	700	77	775	53	530	86	860	ns
R1-R32	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5V	XVIII	S	65	650	65	650	72	720	49	495	79	795	ns
R33-R64	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	XVIII	Q	85	850	85	850	94	940	64	645	104	1040	ns
S1-S32	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	XIX	Q	20	200	20	200	22	220	15	150	24	245	ns
T1-T32	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10V	XX	Q	26	260	26	260	29	290	19	195	32	320	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test conditions	Parameter	Ref test table	Test type 1/ -3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U1-U22	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15V	XXI	Q	15	150	15	150	16	165	11	115	18	185	ns
V1-V22	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	XXII	Q	20	200	20	200	22	220	15	150	24	245	ns
W	Power supply filter capacitance	XXIII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
X1-X22	Input capacitance	XXIV	Q	2	15	2	15	2	15	2	15	2	15	pF

NOTES:

1/ "S" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through G and K through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions H and J shall be performed at 25°, 60° and 100°C using the initial limits for the applicable temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. Use the same test conditions as the maximum class temperature conditions.

4.5.1.2 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition W, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through X shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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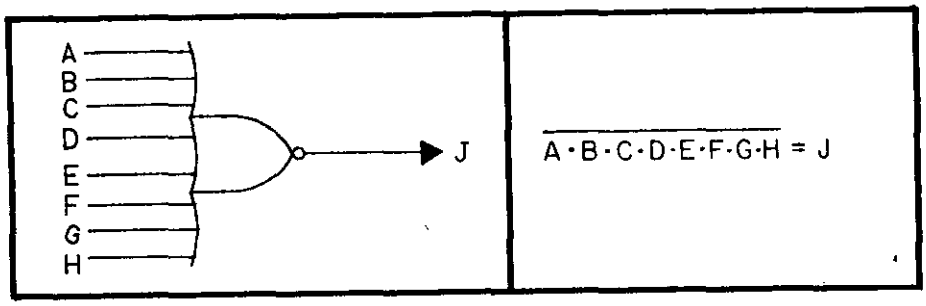
6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	2.4	2.4	mW
Input voltage			
Low level (V_{DD} = 5V)	0.0 min, 1.5 max	0.0 min, 1.5 max	V
(V_{DD} = 15V)	0.0 min, 4.0 max	0.0 min, 4.0 max	V
High level (V_{DD} = 5V)	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
(V_{DD} = 15V)	11 min, V_{DD} max	11 min, V_{DD} max	V
Input currents (V_{DD} = 18V)			
Low level (V_{IL} = 0.0V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18V)	0 min, 200 max	0 min, 200 max	nA
Output drive current (V_{DD} = 5V)			
Low level (V_{OL} = 0.4V)			
I_{OL} at -55°C	---	0.64 min	mA
0° and 25°C	0.51 min	0.51 min	mA
60°C	0.44 min	0.44 min	mA
100°C	---	0.36 min	mA
High level (V_{OH} = 4.6V)			
I_{OH} at -55°C	---	-0.64 min	mA
0° and 25°C	-0.51 min	-0.51 min	mA
60°C	-0.44 min	-0.44 min	mA
100°C	---	-0.36 min	mA

6.5 Module function. The module contains four independent 8 input NAND gates.

6.5.1 Input to output relationship. Each gate shall have the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship for positive logic.



6.5.2 Parallel outputs. Paralleling of output to obtain extended drive and sink capability is not permissible.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.

b. Soldering-iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply (V_{DD}) is off.

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d. All input signals and conditioning voltages must be removed before V_{DD} is disconnected.

e. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the input protection diodes.

f. All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

6.7 Dynamic power dissipation. The dynamic power dissipation can be approximately by the formula:

$$P_{Total} = (CV_{DD}^2 f)N + \text{quiescent power dissipation.}$$

where C = Load capacitance.
V_{DD} = Supply voltage.
f = Operating frequency.
N = Number of outputs.

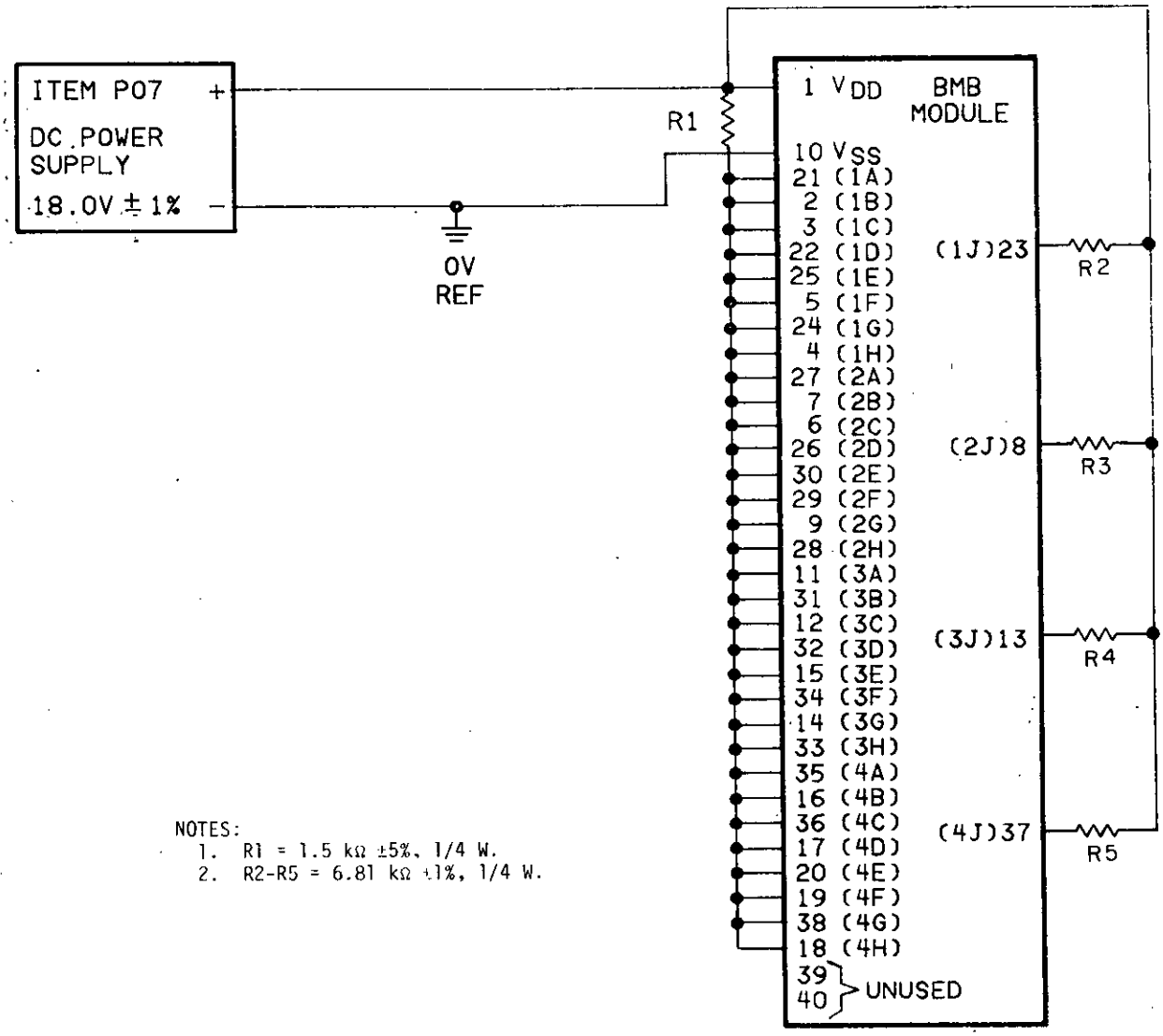
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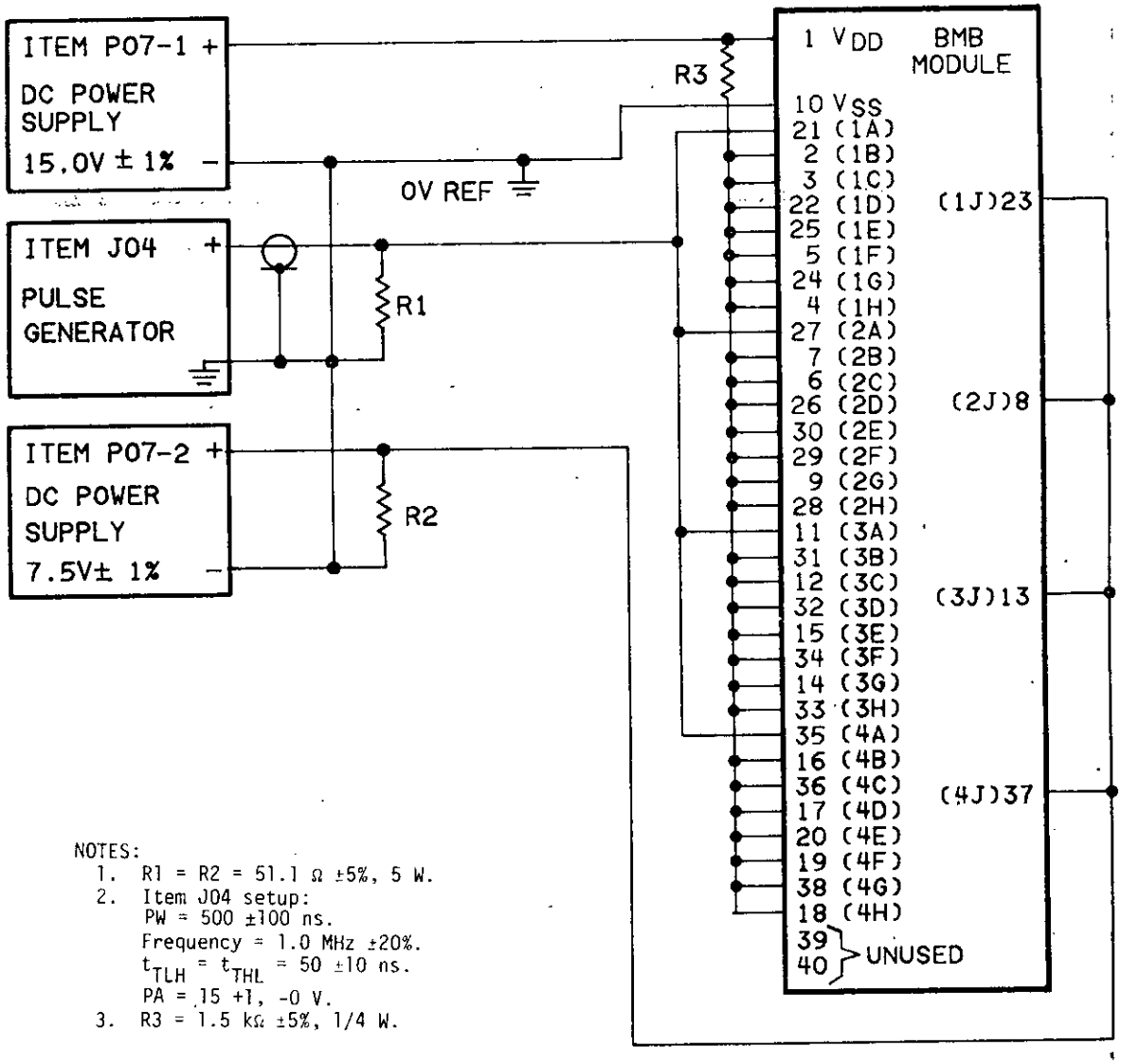
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NOTES:
1. R1 = 1.5 kΩ ±5%, 1/4 W.
2. R2-R5 = 6.81 kΩ ±1%, 1/4 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = R2 = 51.1 Ω ±5%, 5 W.
2. Item J04 setup:
   PW = 500 ±100 ns.
   Frequency = 1.0 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.
   PA = 15 +1, -0 V.
3. R3 = 1.5 kΩ ±5%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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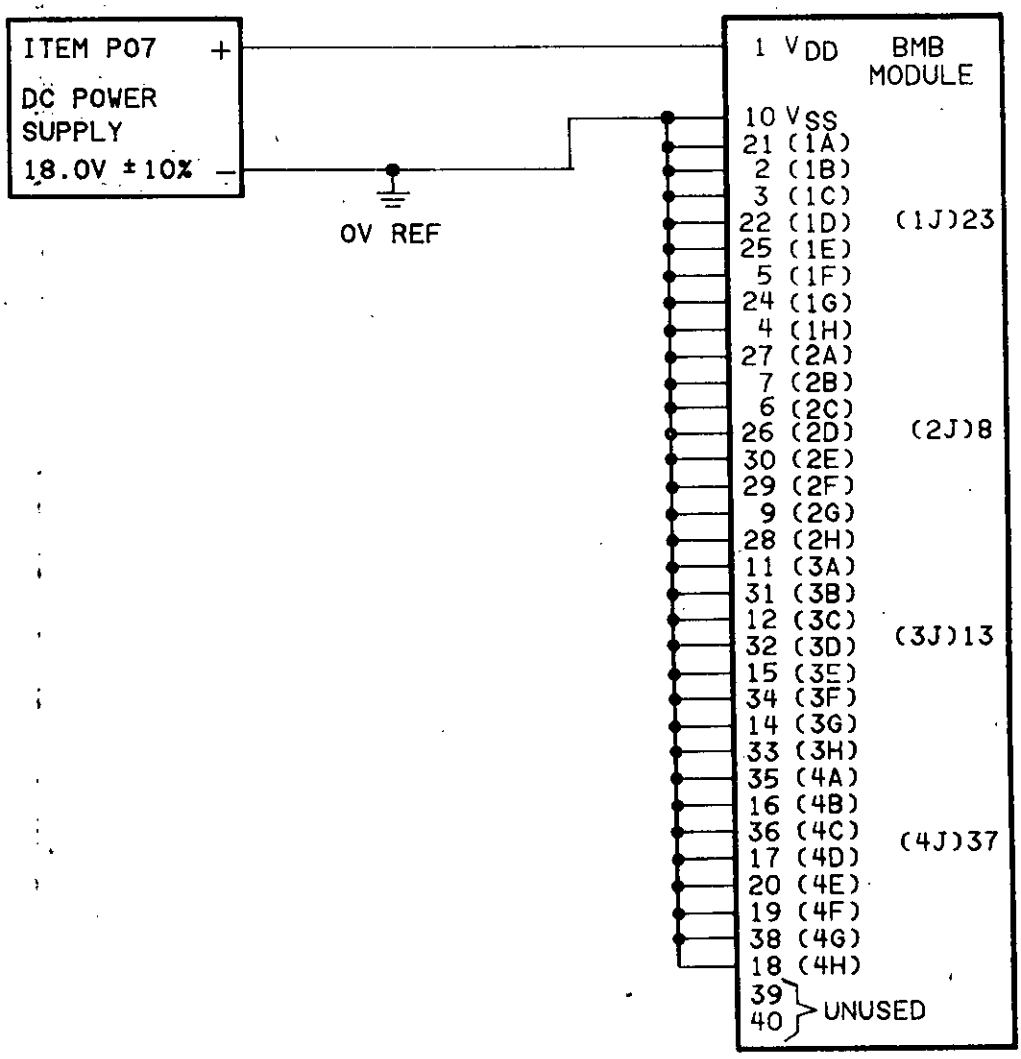


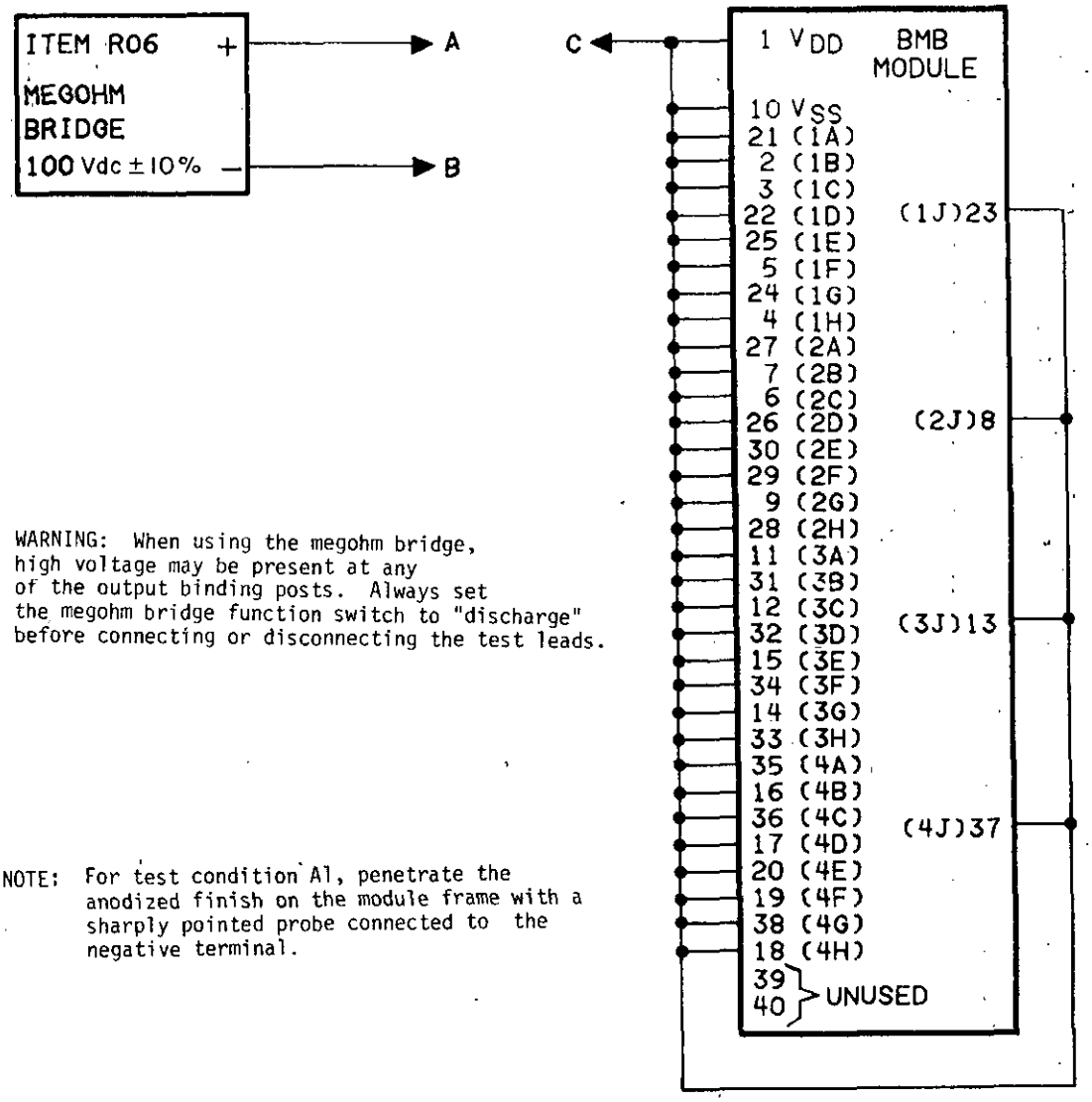
FIGURE 4. Vibration test circuit.

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TABLE III. -Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C, 39, 40	Frame	10	∞	MΩ	5
A2	Isolation	C	39, 40	10	∞	MΩ	5

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WARNING: When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Input clamp diode voltage (negative) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input clamp diode voltage, negative	21	Open	-0.4	-6.0	V	6
B2	1B	2					
B3	1C	3					
B4	1D	22					
B5	1E	25					
B6	1F	5					
B7	1G	24					
B8	1H	4					
B9	2A	27					
B10	2B	7					
B11	2C	6					
B12	2D	26					
B13	2E	30					
B14	2F	29					
B15	2G	9					
B16	2H	28					
B17	3A	11					
B18	3B	31					
B19	3C	12					
B20	3D	32					
B21	3E	15					
B22	3F	34					
B23	3G	14					
B24	3H	33					
B25	4A	35					
B26	4B	16					

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TABLE IV. Input clamp diode voltage (negative) test - Continued.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B27	4C	Input clamp diode voltage, negative	36	Open	-0.4	-6.0	V	6
B28	4D	↓	17	↓	↓	↓	↓	↓
B29	4E		20					
B30	4F		19					
B31	4G		38					
B32	4H		18					

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TABLE V. Input clamp diode voltage (positive) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input clamp diode voltage, positive	Open	21	0.4	1.5	V	6
C2	1B		2				
C3	1C		3				
C4	1D		22				
C5	1E		25				
C6	1F		5				
C7	1G		24				
C8	1H		4				
C9	2A		27				
C10	2B		7				
C11	2C		6				
C12	2D		26				
C13	2E		30				
C14	2F		29				
C15	2G		9				
C16	2H		28				
C17	3A		11				
C18	3B		31				
C19	3C		12				
C20	3D		32				
C21	3E		15				
C22	3F		34				
C23	3G		14				
C24	3H		33				

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TABLE V. Input clamp diode voltage (positive) test - Continued.

Test condition		Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C25	4A	Input clamp diode voltage, positive	Open	35	0.4	1.5	V	6
C26	4B			16				
C27	4C			36				
C28	4D			17				
C29	4E			20				
C30	4F			19				
C31	4G			38				
C32	4H			18				

TABLE VI. Static Output voltage (high-level, I_{OH} = 0) test.

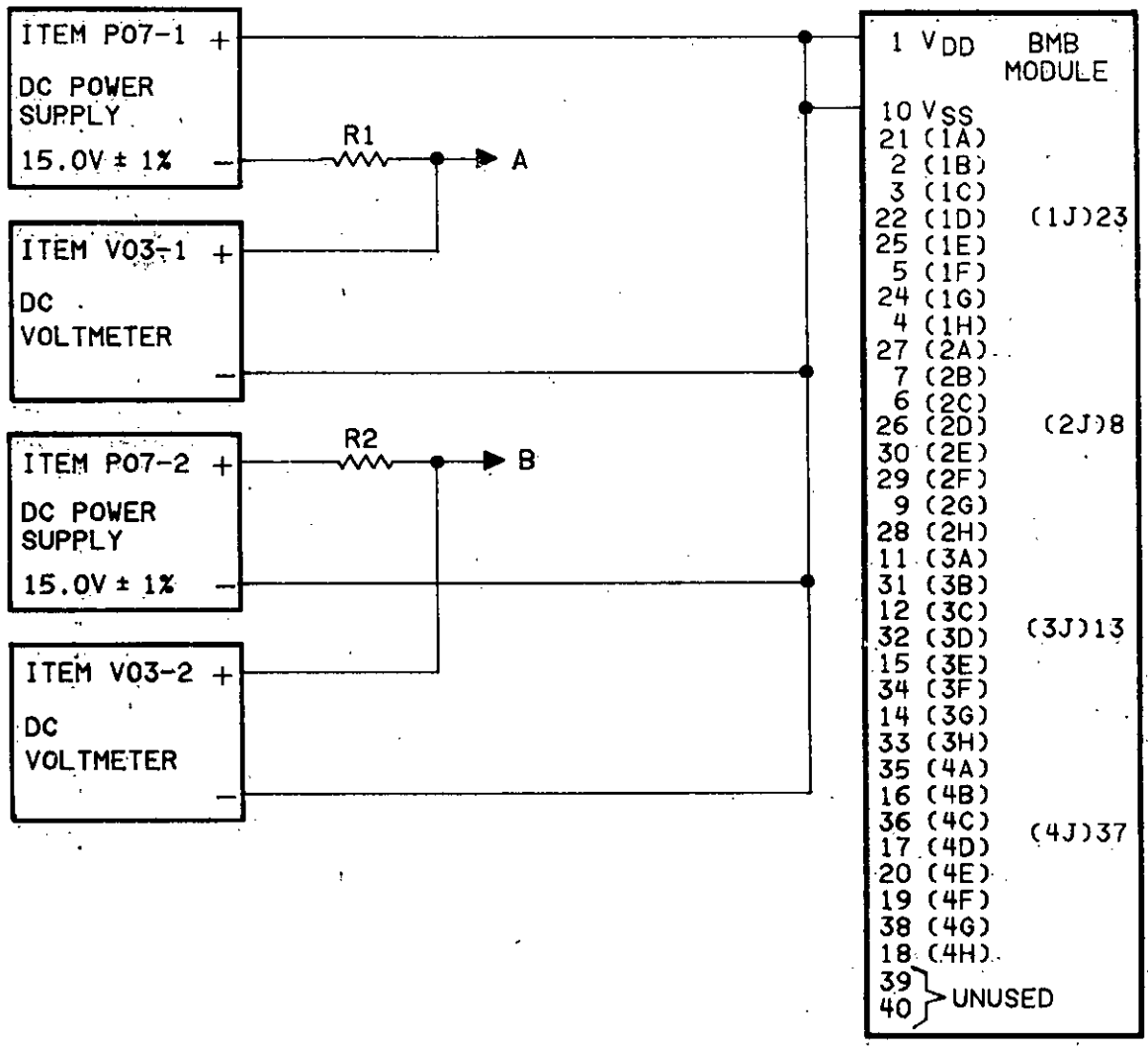
Test condition		Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	1J	Static output voltage, high level (I_{OH} = 0)	23	10	4.95	5.05	V	7
D2	2J		8					
D3	3J		13					
D4	4J		37					

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TABLE VII. Static output voltage (low-level, I_{OL} = 0) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	1J Static output voltage, low level (I_{OL} = 0)	23	1	0.00	0.05	V	7
E2	2J	8					
E3	3J	13					
E4	4J	37					

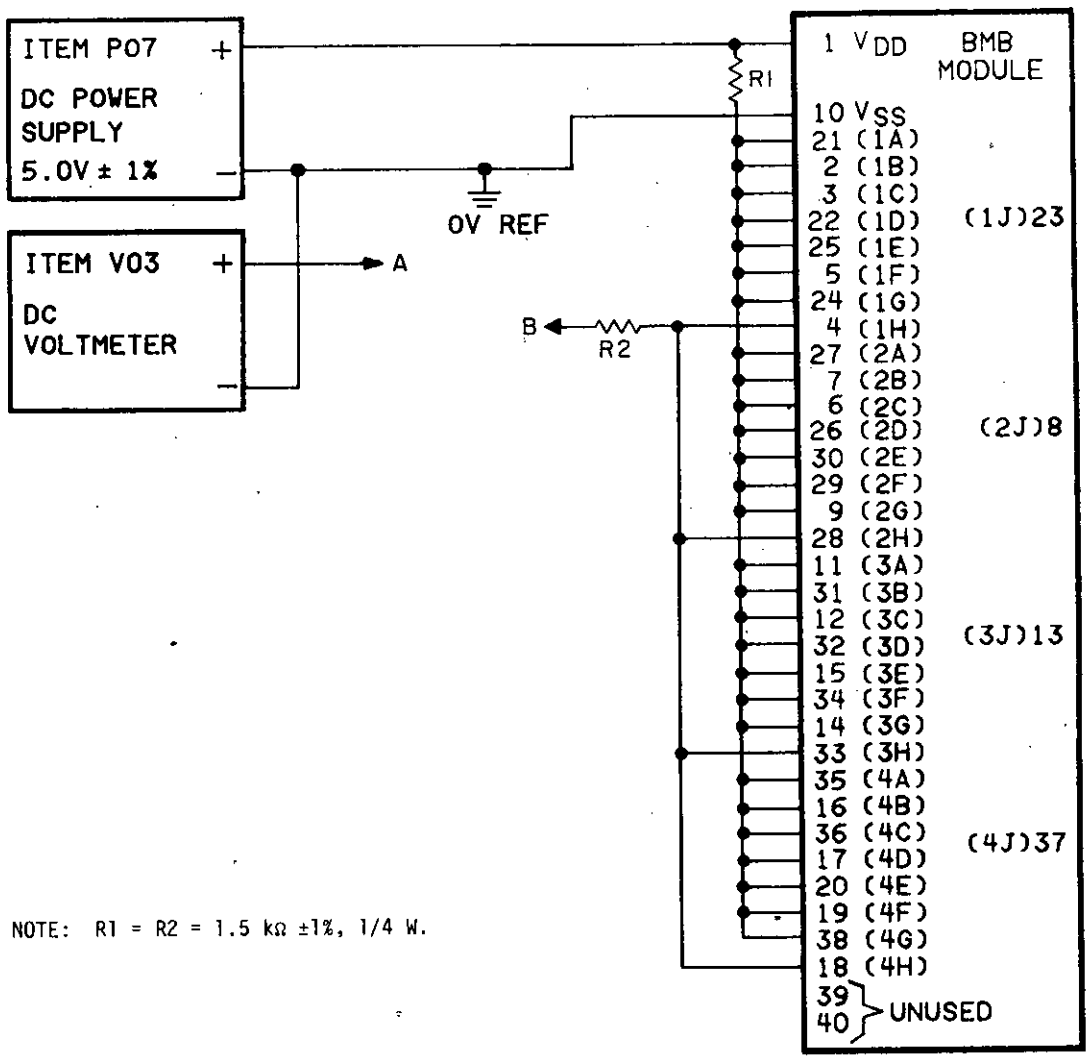
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:06Z
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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items P07-1 and P07-2 and V03-1 and V03-2 are not used simultaneously, only one item P07 and one item V03 is required.

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

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NOTE: R1 = R2 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 7. Static output voltage (high-level, I_{OH} = 0 and low-level, I_{OL} = 0) test circuit for tables VI and VII.

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TABLE VIII. Static output voltage (high-level and low-level) test.

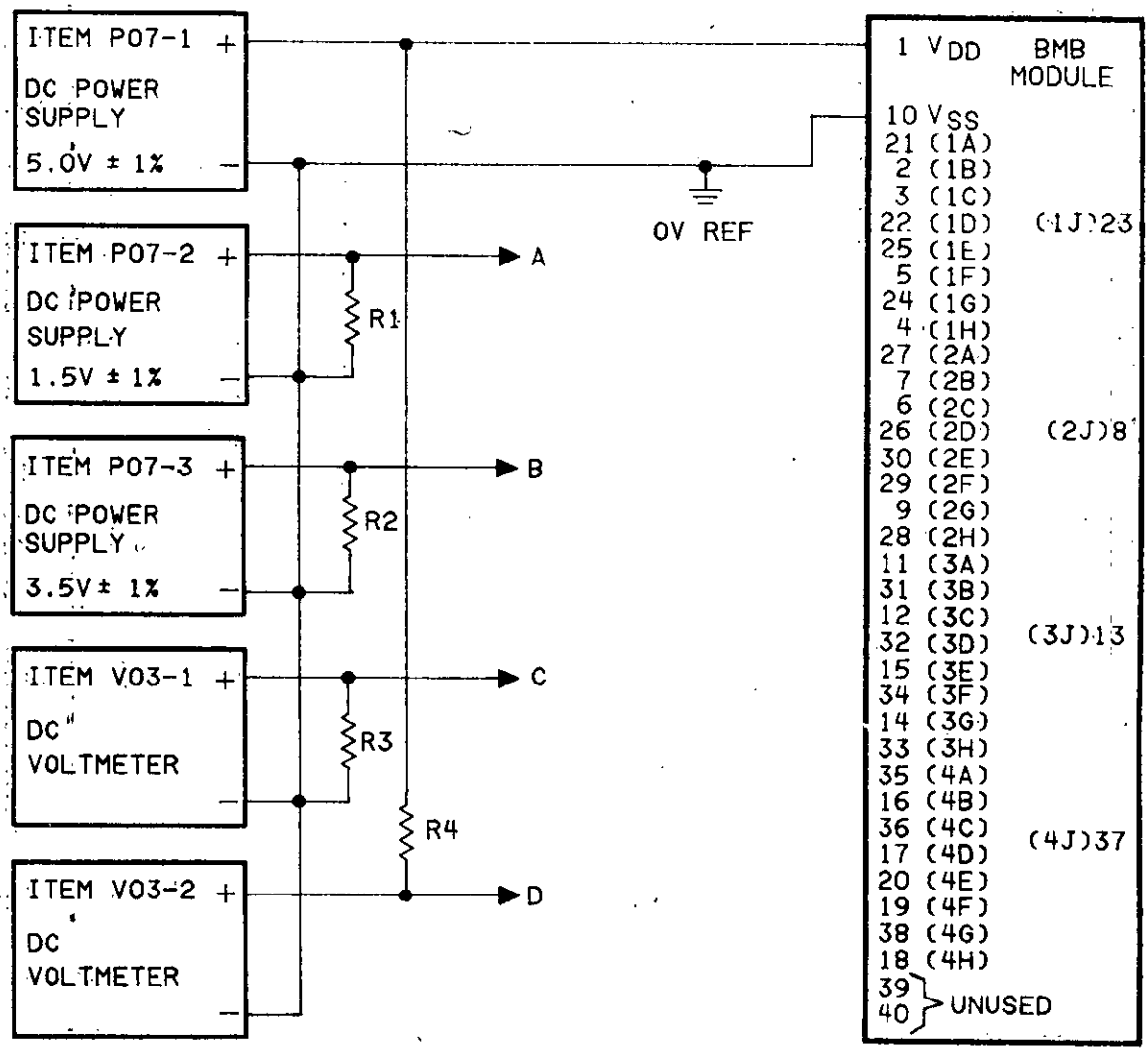
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
F1	1J Static output voltage, high level	2,3,4,5,6,7,9,11,12,14,15, 16,17,18,19,20,21,22,24,25, 26,27,28,29,30,31,32,33,34, 35,36,38	Open	23	Open	4.6	5.0	V	8
F2	2J			8					
F3	3J			13					
F4	4J			37					
F5	1J	2,3,4,5,6,7,9,12,14,15,16, 17,18,19,20,22,24,25,26, 28,29,30,31,32,33,34,36,38	11,21, 27,35	23					
F6	2J			8					
F7	3J			13					
F8	4J			37					
F9	1J	3,4,5,6,9,11,12,14,15,17,18, 19,20,21,22,24,25,26,27,28, 29,30,32,33,34,35,36,38	2,7, 16,31	23					
F10	2J			8					
F11	3J			13					
F12	4J			37					
F13	1J	2,4,5,7,9,11,14,15,16,17,18, 19,20,21,22,24,25,26,27,28, 29,30,31,32,33,34,35,38	3,6, 12,36	23					
F14	2J			8					
F15	3J			13					
F16	4J			37					
F17	1J	2,3,4,5,6,7,9,11,12,14,15, 16,18,19,20,21,24,25,27,28, 29,30,31,33,34,35,36,38	17,22, 26,32	23					
F18	2J			8					
F19	3J			13					
F20	4J			37					
F21	1J	2,3,4,5,6,7,9,11,12,14,16, 17,18,19,21,22,24,26,27,28, 29,31,32,33,34,35,36,38	15,20, 25,30	23					
F22	2J			8					
F23	3J			13					
F24	4J			37					

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TABLE VIII. Static output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
F25	1J	Static output voltage, high level	2,3,4,6,7,9,11,12,14,15,16,17,18,20,21,22,24,25,26,27,28,30,31,32,33,35,36,38	5,19,29,34	23	Open	4.6	5.0	V	8
F26	2J				8					
F27	3J				13					
F28	4J				37					
F29	1J		2,3,4,5,6,7,11,12,15,16,17,18,19,20,21,22,25,26,27,28,29,30,31,32,33,34,35,36	9,14,24,38	23					
F30	2J				8					
F31	3J				13					
F32	4J				37					
F33	1J		2,3,5,6,7,9,11,12,14,15,16,17,19,20,21,22,24,25,26,27,29,30,31,32,34,35,36,38	4,18,28,33	23					
F34	2J				8					
F35	3J				13					
F36	4J				37					
F37	1J	Static output voltage, low level	Open	2,3,4,5,6,7,9,11,12,14,15,16,17,18,19,20,21,22,24,25,26,27,28,29,30,31,32,33,35,36,38	Open	23	0.0	0.4		
F38	2J					8				
F39	3J					13				
F40	4J					37				

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NOTES:
1. R1 = R2 = 51.1 Ω ±5%, 1/2 W.
2. R3 = 18.2 kΩ ±1%, 1/4 W @ -55°C
   = 23.2 kΩ ±1%, 1/4 W @ 0° and 25°C
   = 27.4 kΩ ±1%, 1/4 W @ 60°C
   = 32.4 kΩ ±1%, 1/4 W @ 100°C
3. R4 = 7.15 kΩ ±1%, 1/4 W @ -55°C
   = 9.09 kΩ ±1%, 1/4 W @ 0° and 25°C
   = 10.5 kΩ ±1%, 1/4 W @ 60°C
   = 12.7 kΩ ±1%, 1/4 W @ 100°C

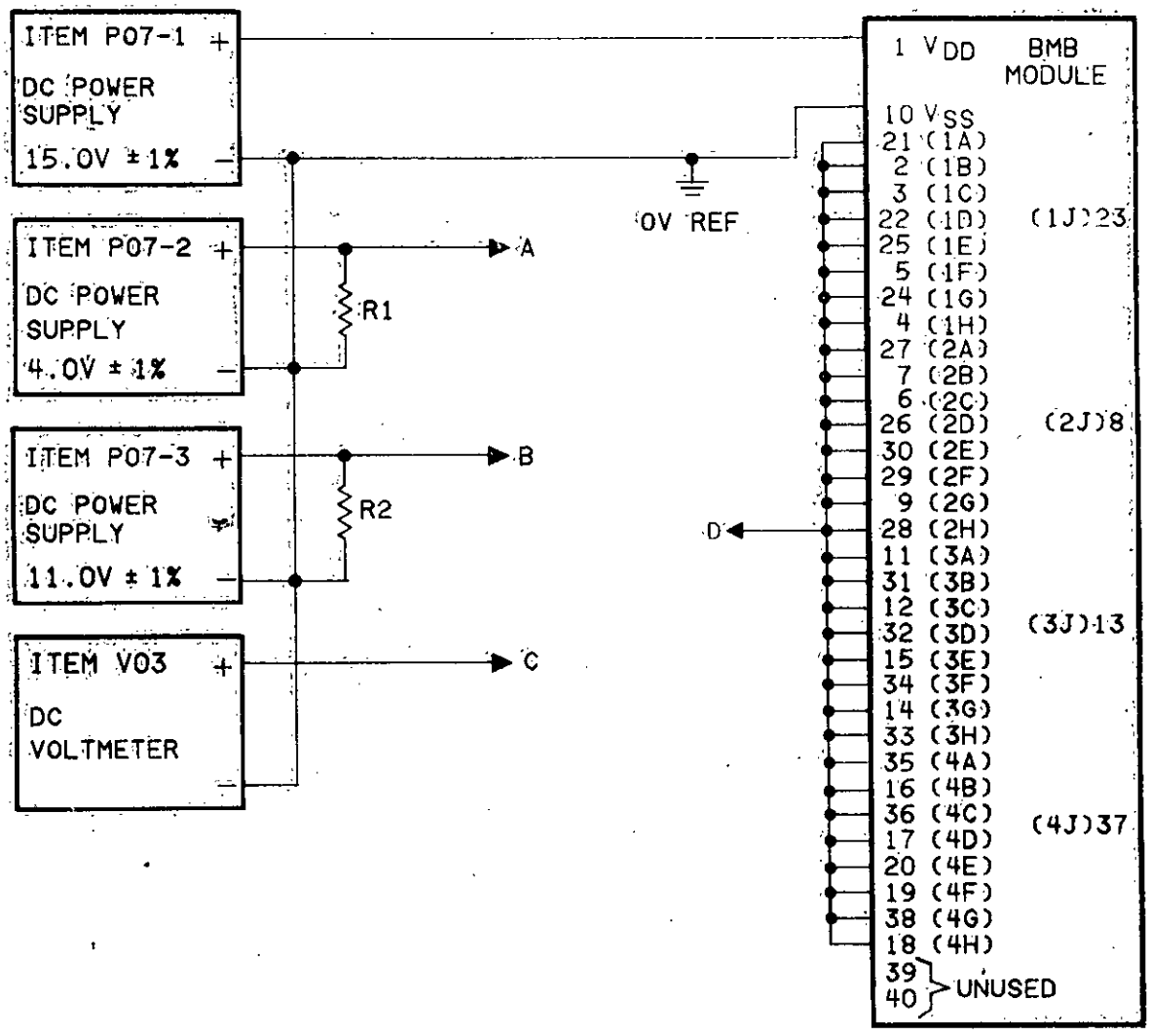
FIGURE 8. Static output voltage (high-level and low-level) test circuit for table VIII.

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TABLE IX. Static output voltage (high-level and low-level, V_{DD} = 15V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1J Static output voltage, high level, V_{DD} = 15V	D	Open	23	13.5	15.0	V	9
G2	2J	↓	↓	8	↓	↓	↓	↓
G3	3J	↓	↓	13	↓	↓	↓	↓
G4	4J	↓	↓	37	↓	↓	↓	↓
G5	1J Static output voltage, low level, V_{DD} = 15V	Open	D	23	0.0	1.5	↓	↓
G6	2J	↓	↓	8	↓	↓	↓	↓
G7	3J	↓	↓	13	↓	↓	↓	↓
G8	4J	↓	↓	37	↓	↓	↓	↓

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NOTES:
1. R1 = 51.1 Ω ±5%, 1/2 W.
2. R2 = 576 Ω ±5%, 1/4 W.

FIGURE 9. Static output voltage (high-level and low-level, V_{DD} = 15 V) test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:06Z
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TABLE X. Input current (high-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
H1	1A Input current, high level	2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11, 31,12,32,15,34,14,33,35,16,36,17,20,19, 38,18	21	0	200	nA	10
H2	1B	21,3,22,25,5,24,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	2				
H3	1C	21,2,22,25,5,24,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	3				
H4	1D	21,2,3,25,5,24,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	22				
H5	1E	21,2,3,22,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,36,17, 20,19,38,18	25				
H6	1F	21,2,3,22,25,24,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	5				
H7	1G	21,2,3,22,25,5,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	24				
H8	1H	21,2,3,22,25,5,24,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	4				
H9	2A	21,2,3,22,25,5,24,4,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	27				
H10	2B	21,2,3,22,25,5,24,4,27,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	7				

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TABLE X. Input current (high-level) test - Continued.

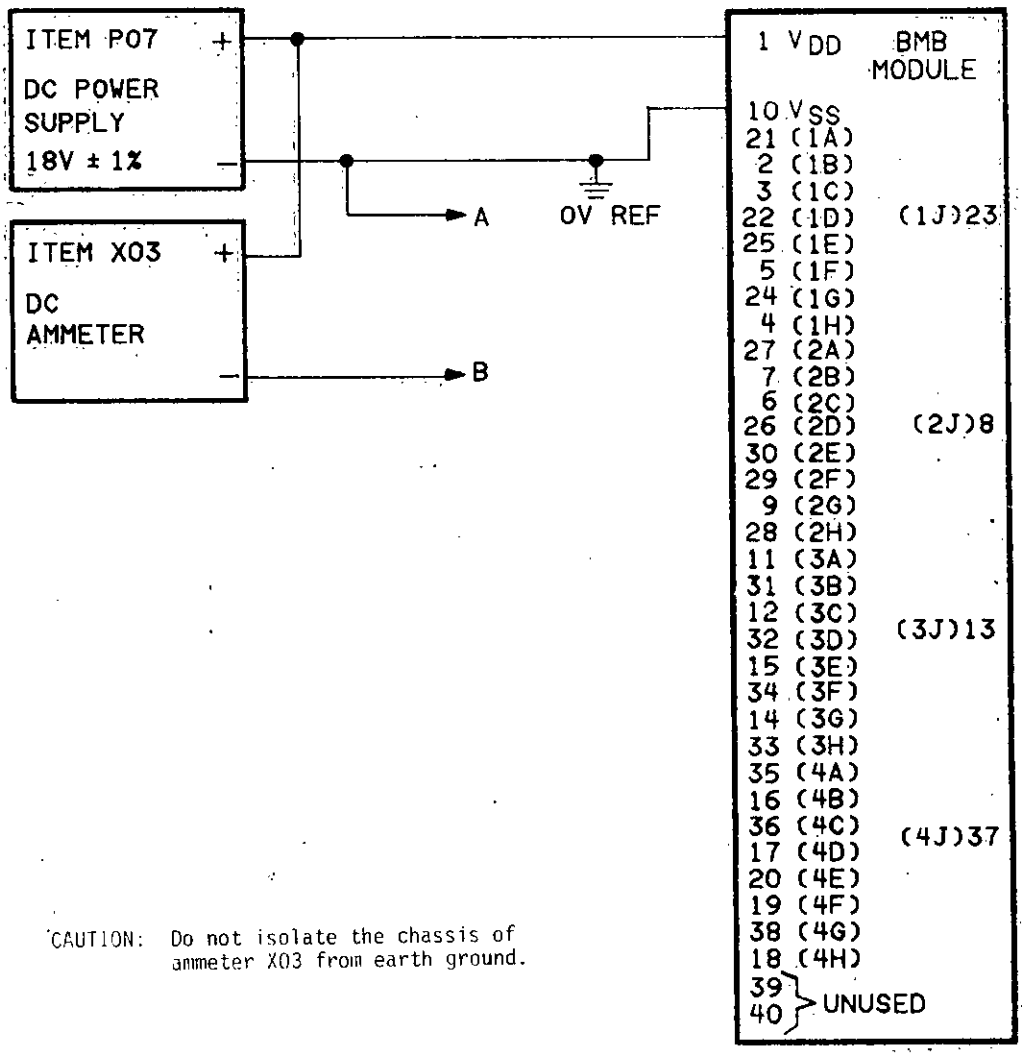
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
H11	2C Input current, high level	21,2,3,22,25,5,24,4,27,7,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	6	0	200	nA	10
H12	2D	21,2,3,22,25,5,24,4,27,7,6,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	26				
H13	2E	21,2,3,22,25,5,24,4,27,7,6,26,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	30				
H14	2F	21,2,3,22,25,5,24,4,27,7,6,26,30,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	29				
H15	2G	21,2,3,22,25,5,24,4,27,7,6,26,30,29,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	9				
H16	2H	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	28				
H17	3A	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	11				
H18	3B	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	31				
H19	3C	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,32,15,34,14,33,35,16,36,17,20, 19,38,18	12				
H20	3D	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,15,34,14,33,35,16,36,17,20, 19,38,18	32				

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TABLE X. Input current (high-level) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
H21	3E Input current, high level	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,34,14,33,35,16,36,17,20, 19,38,18	15	0	200	nA	10
H22	3F	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,14,33,35,16,36,17,20, 19,38,18	34				
H23	3G	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,33,35,16,36,17,20, 19,38,18	14				
H24	3H	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,35,16,36,17,20, 19,38,18	33				
H25	4A	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,16,36,17,20,19, 38,18	35				
H26	4B	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,14,34,33,35,36,17,20, 19,38,18	16				
H27	4C	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,17,20, 19,38,18	36				
H28	4D	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9 28,11,31,12,32,15,34,14,33,35,16,36,20, 19,38,18	17				
H29	4E	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,36,17, 19,38,18	20				
H30	4F	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,36,17, 20,38,18	19				
H31	4G	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,36,17, 20,19,18	38				
H32	4H	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,36,17, 20,19,38	18				

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CAUTION: Do not isolate the chassis of ammeter X03 from earth ground.

FIGURE 10. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
J1	1A Input current, low level	2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11, 31,12,32,15,34,14,33,35,16,36,17,20,19, 38,18	21	0	-200	nA	11
J2	1B	21,3,22,25,5,24,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	2				
J3	1C	21,2,22,25,5,24,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,36,16,17,20, 19,38,18	3				
J4	1D	21,2,3,25,5,24,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	22				
J5	1E	21,2,3,22,5,24,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17, 20,19,38,18	25				
J6	1F	21,2,3,22,25,24,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	5				
J7	1G	21,2,3,22,25,5,4,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	24				
J8	1H	21,2,3,22,25,5,24,27,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	4				
J9	2A	21,2,3,22,25,5,24,4,7,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	27				
J10	2B	21,2,3,22,25,5,24,4,27,6,26,30,29,9,28, 11,31,12,32,15,34,14,33,35,16,36,17,20, 19,38,18	7				

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TABLE XI. Input current (low-level) test - Continued.

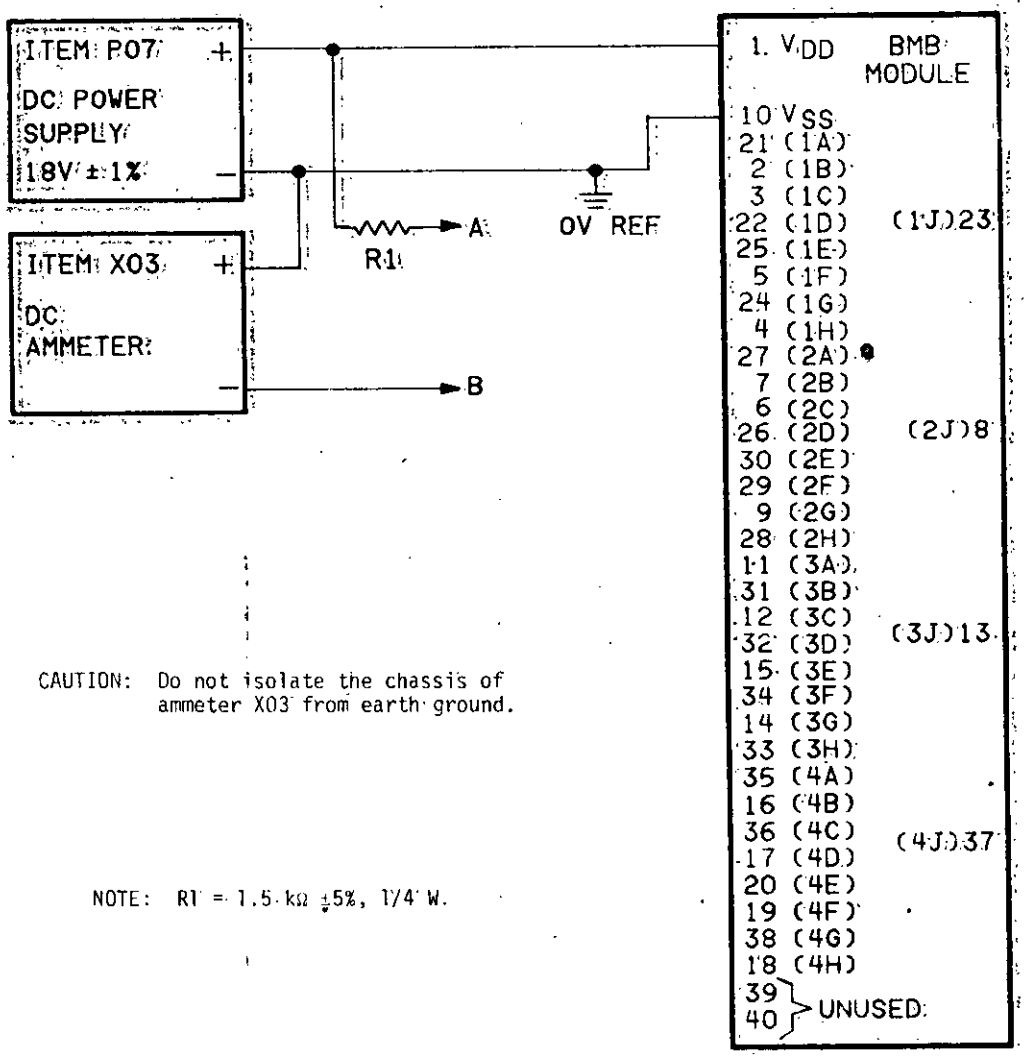
Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
J11	2C Input current, low level	21,2,3,22,25,5,24,4,27,7,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	6	0	-200	mA	11
J12	2D	21,2,3,22,25,5,24,4,27,7,6,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	26				
J13	2E	21,2,3,22,25,5,24,4,27,7,6,26,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	30				
J14	2F	21,2,3,22,25,5,24,4,27,7,6,26,30,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	29				
J15	2G	21,2,3,22,25,5,24,4,27,7,6,26,30,29,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	9				
J16	2H	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	28				
J17	3A	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	11				
J18	3B	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,12,32,15,34,14,33,35,16,36,17,20,19,38,18	31				
J19	3C	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,32,15,34,14,33,35,16,36,17,20,19,38,18	12				
J20	3D	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,15,34,14,33,35,16,36,17,20,19,38,18	32				

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TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J21	3E Input current, low level	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,34,14,33,35,16,36,17,20, 19,38,18	15	0	-200	nA	11
J22	3F	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,14,33,35,16,36,17,20, 19,38,18	34				
J23	3G	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,33,35,16,36,17,20, 19,38,18	14				
J24	3H	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,35,16,36,17,20, 19,38,18	33				
J25	4A	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,16,36,17,20, 19,38,18	35				
J26	4B	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,36,17,20, 19,38,18	16				
J27	4C	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,17,20, 19,38,18	36				
J28	4D	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9 28,11,31,12,32,15,34,14,33,35,16,36,20, 19,38,18	17				
J29	4E	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,36,17, 19,38,18	20				
J30	4F	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,36,17, 20,38,18	19				
J31	4G	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,36,17, 20,19,18	38				
J32	4H	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9, 28,11,31,12,32,15,34,14,33,35,16,36,17, 20,19,38	18				

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CAUTION: Do not isolate the chassis of ammeter X03 from earth ground.

NOTE: R1 = 1.5 kΩ ±5%, 1/4 W.

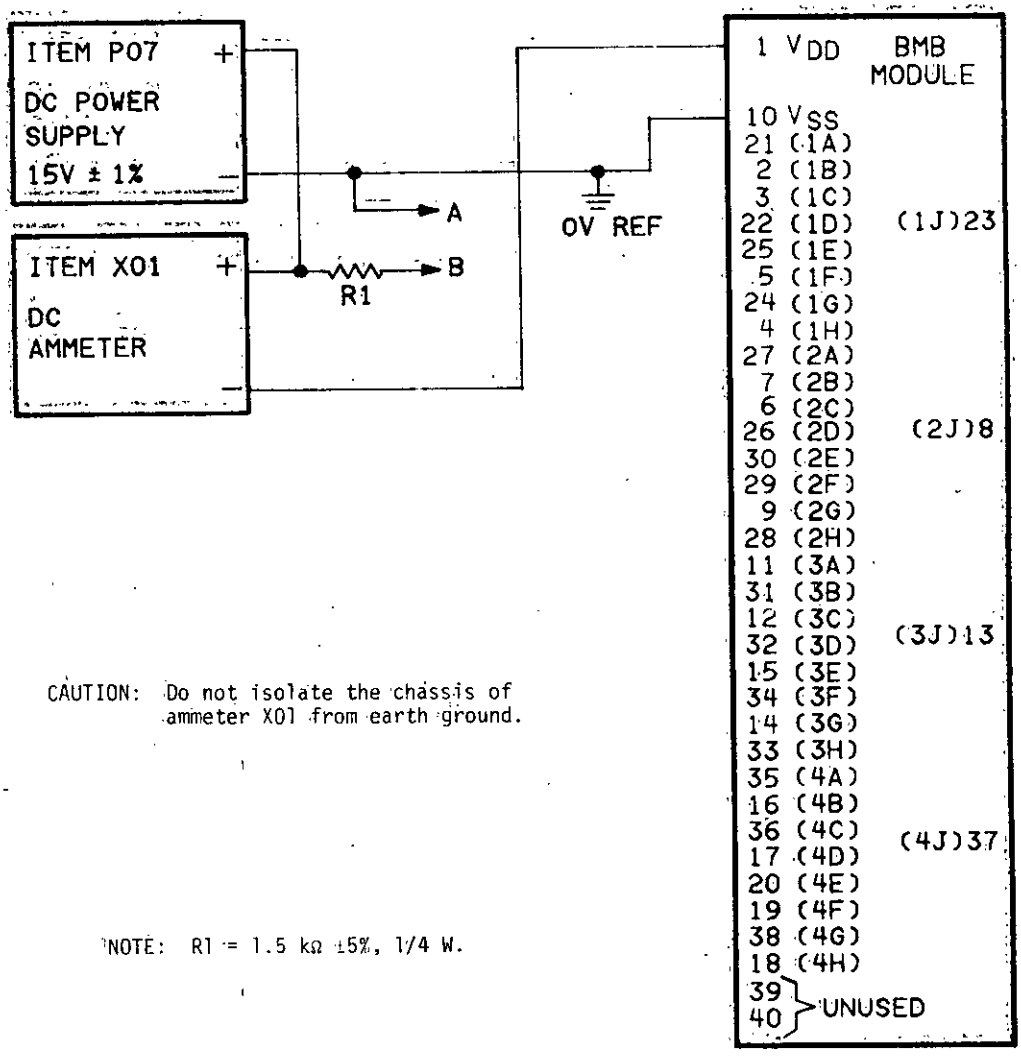
FIGURE 11. Input current (low-level) test circuit for table XI.

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TABLE XII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Power supply current, low level	Open	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	0	8	µA	12
K2	Power supply current, high level	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	Open	0	8	µA	12

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: R1 = 1.5 kΩ ±5%, 1/4 W.

FIGURE 12. Power supply current (low-level and high-level) test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH}, V_{DD} = 5V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1J Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5V	27, 11, 35	21	23	20	200	ns	13,15
L2	2J	21, 11, 35	27	8	↓	↓	↓	↓
L3	3J	21, 27, 35	11	13				
L4	4J	21, 27, 11	35	37				
L5	1J Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5V	27, 11, 35	21	23	65	680		
L6	2J	21, 11, 35	27	8	↓	↓		
L7	3J	21, 27, 35	11	13				
L8	4J	21, 27, 11	35	37				

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TABLE XIV. Transition time (t_{THL}, V_{DD} = 5V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	1J Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5V	27, 11, 35	21	23	20	200	ns	13,15
M2	2J	21, 11, 35	27	8				
M3	3J	21, 27, 35	11	13				
M4	4J	21, 27, 11	35	37				
M5	1J Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5V	27, 11, 35	21	23	65	680		
M6	2J	21, 11, 35	27	8				
M7	3J	21, 27, 35	11	13				
M8	4J	21, 27, 11	35	37				

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TABLE XV. Transition time (t_{TLH} and t_{THL}, V_{DD} = 10V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1J Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10V	27, 11, 35	21	23	10	100	ns	13, 15
N2	2J	21, 11, 35	27	8				
N3	3J	21, 27, 35	11	13				
N4	4J	21, 27, 11	35	37				
N5	1J Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10V	27, 11, 35	21	23				
N6	2J	21, 11, 35	27	8				
N7	3J	21, 27, 35	11	13				
N8	4J	21, 27, 11	35	37				

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TABLE XVI. Transition time (t_{TLH} and t_{THL}; V_{DD} = 15V) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
P1	1J Transition time (t_{TLH}); C_L = 50 pF; V_{DD} = 15V	27, 11, 35	21	23	8	80	ns	13,15
P2	2J	21, 11, 35	27	8				
P3	3J	21, 27, 35	11	13				
P4	4J	21, 27, 11	35	37				
P5	1J Transition time (t_{THL}); C_L = 50 pF; V_{DD} = 15V	27, 11, 35	21	23				
P6	2J	21, 11, 35	27	8				
P7	3J	21, 27, 35	11	13				
P8	4J	21, 27, 11	35	37				

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TABLE XVII. Propagation delay time (t_{PLH}, V_{DD} = 5V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	1A to 1J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	2,3,22,25,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	21	23	50	500	ns	14,15
Q2	1B to 1J	21,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	2					
Q3	1C to 1J	21,2,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	3					
Q4	1D to 1J	21,2,3,25,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	22					
Q5	1E to 1J	21,2,3,22,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	25					
Q6	1F to 1J	21,2,3,22,25,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	5					
Q7	1G to 1J	21,2,3,22,25,5,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	24					
Q8	1H to 1J	21,2,3,22,25,5,24,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	4					
Q9	2A to 2J	21,2,3,22,25,5,24,4,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	27	8				
Q10	2B to 2J	21,2,3,22,25,5,24,4,27,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	7					
Q11	2C to 2J	21,2,3,22,25,5,24,4,27,7,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	6					
Q12	2D to 2J	21,2,3,22,25,5,24,4,27,7,6,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	26					

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TABLE XVII. Transition time (t_{PLH}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
Q13	2E to 2J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	21,2,3,22,25,5,24,4,27,7,6,26,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	30	8	50	500	ns	14, 15
Q14	2F to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	29					
Q15	2G to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	9					
Q16	2H to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	28					
Q17	3A to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	11	13				
Q18	3B to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,12,32,15,34,14,33,35,16,36,17,20,19,38,18	31					
Q19	3C to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,32,15,34,14,33,35,16,36,17,20,19,38,18	12					
Q20	3D to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,15,34,14,33,35,16,36,17,20,19,38,18	32					
Q21	3E to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,34,14,33,35,16,36,17,20,19,38,18	15					
Q22	3F to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,14,33,35,16,36,17,20,19,38,18	34					
Q23	3G to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,33,35,16,36,17,20,19,38,18	14					
Q24	3H to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,35,16,36,17,20,19,38,18	33					

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TABLE XVII. Propagation delay time (t_{PLH}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q25	4A to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 16,36,17,20,19,38,18	35	37	50	500	ns	14,15
Q26	4B to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,36,17,20,19,38,18	16					
Q27	4C to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,17,20,19,38,18	36					
Q28	4D to 4J	21,2,3,22,25,5,24,4,27,7,6,26, 30,29,9,28,11,31,12,32,15,34,14, 33,35,16,36,20,19,38,18	17					
Q29	4E to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,19,38,18	20					
Q30	4F to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,38,18	19					
Q31	4G to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,18	38					
Q32	4H to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38	18					
Q33	1A to 1J Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	2,3,22,25,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	21	23	70	700		
Q34	1B to 1J	21,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	2					
Q35	1C to 1J	21,2,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	3					
Q36	1D to 1J	21,2,3,25,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	22					

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TABLE XVII. Propagation delay time (t_{PLH}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q37	1E to 1J Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	21,2,3,22,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	25	23	70	700	ns	14,15
Q38	1F to 1J	21,2,3,22,25,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	5					
Q39	1G to 1J	21,2,3,22,25,5,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	24					
Q40	1H to 1J	21,2,3,22,25,5,24,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	4					
Q41	2A to 2J	21,2,3,22,25,5,24,4,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	27	8				
Q42	2B to 2J	21,2,3,22,25,5,24,4,27,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	7					
Q43	2C to 2J	21,2,3,22,25,5,24,4,27,7,26,30, 29,9,28,11,31,12,32,15,34,14, 33,35,16,36,17,20,19,38,18	6					
Q44	2D to 2J	21,2,3,22,25,5,24,4,27,7,6,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	26					
Q45	2E to 2J	21,2,3,22,25,5,24,4,27,7,6,26,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	30					
Q46	2F to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	29					
Q47	2G to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	9					
Q48	2H to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	28					

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TABLE XVII. Propagation delay time (t_{PLH}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q49	3A to 3J Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	11	13	70	700	ns	14,15
Q50	3B to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	31					
Q51	3C to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,32,15,34,14,33,35, 16,36,17,20,19,38,18	12					
Q52	3D to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,15,34,14,33,35, 16,36,17,20,19,38,18	32					
Q53	3E to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,34,14,33, 35,16,36,17,20,19,38,18	15					
Q54	3F to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,14,33,35, 16,36,17,20,19,38,18	34					
Q55	3G to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,33,35, 16,36,17,20,19,38,18	14					
Q56	3H to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,35, 16,36,17,20,19,38,18	33					
Q57	4A to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 16,36,17,20,19,38,18	35	37				
Q58	4B to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,36,17,20,19,38,18	16					
Q59	4C to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,17,20,19,38,18	36					
Q60	4D to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,20,19,38,18	17					

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TABLE XVII. Propagation delay time (t_{PLH}, V_{DD} = 5V) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q61	4E to 4J	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,19,38,18	20	37	70	700	ns	14,15
Q62	4F to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,38,18	19					
Q63	4G to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,18	38					
Q64	4H to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38	18					

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TABLE XVIII. Propagation delay time (t_{PHL}, V_{DD} = 5V) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
R1	1A to 1J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5V	2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	21	23	65	650	ns	14,15
R2	1B to 1J	21,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	2					
R3	1C to 1J	21,2,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	3					
R4	1D to 1J	21,2,3,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	22					
R5	1E to 1J	21,2,3,22,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	25					
R6	1F to 1J	21,2,3,22,25,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	5					
R7	1G to 1J	21,2,3,22,25,5,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	24					
R8	1H to 1J	21,2,3,22,25,5,24,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	4					
R9	2A to 2J	21,2,3,22,25,5,24,4,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	27	8				
R10	2B to 2J	21,2,3,22,25,5,24,4,27,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	7					
R11	2C to 2J	21,2,3,22,25,5,24,4,27,7,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	6					
R12	2D to 2J	21,2,3,22,25,5,24,4,27,7,6,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	26					

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TABLE XVIII. Transition time (t_{PHL}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
R13	2E to 2J	21,2,3,22,25,5,24,4,27,7,6,26,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	30	8	65	650	ns	14,15
R14	2F to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	29					
R15	2G to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	9					
R16	2H to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	28					
R17	3A to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	11	13				
R18	3B to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,12,32,15,34,14,33,35,16,36,17,20,19,38,18	31					
R19	3C to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,32,15,34,14,33,35,16,36,17,20,19,38,18	12					
R20	3D to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,15,34,14,33,35,16,36,17,20,19,38,18	32					
R21	3E to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,34,14,33,35,16,36,17,20,19,38,18	15					
R22	3F to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,14,33,35,16,36,17,20,19,38,18	34					
R23	3G to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,33,35,16,36,17,20,19,38,18	14					
R24	3H to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,35,16,36,17,20,19,38,18	33					

Parameter: Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5V

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TABLE XVIII. Propagation delay time (t_{PHL}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
R25	4A to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5V	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 16,36,17,20,19,38,18	35	37	65	650	ns	14,15
R26	4B to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,36,17,20,19,38,18	16					
R27	4C to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,17,20,19,38,18	36					
R28	4D to 4J	21,2,3,22,25,5,24,4,27,7,6,26, 30,29,9,28,11,31,12,32,15,34,14, 33,35,16,36,20,19,38,18	17					
R29	4E to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,19,38,18	20					
R30	4F to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,38,18	19					
R31	4G to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,18	38					
R32	4H to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38	18					
R33	1A to 1J Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	2,3,22,25,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	21	23	85	850		
R34	1B to 1J	21,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	2					
R35	1C to 1J	21,2,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	3					
R36	1D to 1J	21,2,3,25,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	22					

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TABLE XVIII. Propagation delay time (t_{PHL}, V_{DD} = 5V) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R37	1E to 1J	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	21,2,3,22,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	25	23	85	850	ns	14,15
R38	1F to 1J		21,2,3,22,25,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	5					
R39	1G to 1J		21,2,3,22,25,5,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	24					
R40	1H to 1J		21,2,3,22,25,5,24,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	4					
R41	2A to 2J		21,2,3,22,25,5,24,4,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	27	8				
R42	2B to 2J		21,2,3,22,25,5,24,4,27,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	7					
R43	2C to 2J		21,2,3,22,25,5,24,4,27,7,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	6					
R44	2D to 2J		21,2,3,22,25,5,24,4,27,7,6,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	26					
R45	2E to 2J		21,2,3,22,25,5,24,4,27,7,6,26,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	30					
R46	2F to 2J		21,2,3,22,25,5,24,4,27,7,6,26,30,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	29					
R47	2G to 2J		21,2,3,22,25,5,24,4,27,7,6,26,30,29,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	9					
R48	2H to 2J		21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	28					

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TABLE XVIII. Propagation delay time (t_{PHL}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R49	3A to 3J Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	11	13	85	850	ns	14,15
R50	3B to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	31					
R51	3C to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,32,15,34,14,33,35, 16,36,17,20,19,38,18	12					
R52	3D to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,15,34,14,33,35, 16,36,17,20,19,38,18	32					
R53	3E to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,34,14,33, 35,16,36,17,20,19,38,18	15					
R54	3F to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,14,33,35, 16,36,17,20,19,38,18	34					
R55	3G to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,33,35, 16,36,17,20,19,38,18	14					
R56	3H to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,35, 16,36,17,20,19,38,18	33					
R57	4A to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 16,36,17,20,19,38,18	35	37				
R58	4B to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,36,17,20,19,38,18	16					
R59	4C to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,17,20,19,38,18	36					

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TABLE XVIII. Propagation delay time (t_{PHL}, V_{DD} = 5V) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R60	4D to 4J	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,32,31,12,15,34,14,33, 35,16,36,20,19,38,18	17	37	85	850	ns	14,15
R61	4E to 4J	↓	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,19,38,18	20	↓	↓	↓	↓	↓
R62	4F to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,38,18	19					
R63	4G to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,18	38					
R64	4H to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38	18					

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TABLE XIX. Propagation delay time (t_{PLH}, V_{DD} = 10V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	1A to 1J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	2,3,22,25,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	21	23	20	200	ns	14,15
S2	1B to 1J	21,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	2					
S3	1C to 1J	21,2,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	3					
S4	1D to 1J	21,2,3,25,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	22					
S5	1E to 1J	21,2,3,22,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	25					
S6	1F to 1J	21,2,3,22,25,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	5					
S7	1G to 1J	21,2,3,22,25,5,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	24					
S8	1H to 1J	21,2,3,22,25,5,24,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	4					
S9	2A to 2J	21,2,3,22,25,5,24,4,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	27	8				
S10	2B to 2J	21,2,3,22,25,5,24,4,27,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	7					
S11	2C to 2J	21,2,3,22,25,5,24,4,27,7,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	6					
S12	2D to 2J	21,2,3,22,25,5,24,4,27,7,6,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	26					

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TABLE XIX. Propagation delay time (t_{PLH}, V_{DD} = 10V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S13	2E to 2J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	21,2,3,22,25,5,24,4,27,7,6,26,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	30	8	20	200	ns	14,15
S14	2F to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	29					
S15	2G to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	9					
S16	2H to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	28					
S17	3A to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	11	13				
S18	3B to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,12,32,15,34,14,33,35,16,36,17,20,19,38,18	31					
S19	3C to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,32,15,34,14,33,35,16,36,17,20,19,38,18	12					
S20	3D to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,15,34,14,33,35,16,36,17,20,19,38,18	32					
S21	3E to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,34,14,33,35,16,36,17,20,19,38,18	15					
S22	3F to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,14,33,35,16,36,17,20,19,38,18	34					
S23	3G to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,33,35,16,36,17,20,19,38,18	14					
S24	3H to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,35,16,36,17,20,19,38,18	33					

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TABLE XIX. Propagation delay time (t_{PLH}, V_{DD} = 10V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S25	4A to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 16,36,17,20,19,38,18	35	37	20	200	ns	14,15
S26	4B to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,36,17,20,19,38,18	16					
S27	4C to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,17,20,19,38,18	36					
S28	4D to 4J	21,2,3,22,25,5,24,4,27,7,6,26, 30,29,9,28,11,31,12,32,15,34,14, 33,35,16,36,20,19,38,18	17					
S29	4E to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,19,38,18	20					
S30	4F to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,38,18	19					
S31	4G to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,18	38					
S32	4H to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38	18					

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TABLE XX. Propagation delay time (t_{PHL}, V_{DD} = 10V) test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
T1	1A to 1J	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10V	2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	21	23	26	260	ns	14,15
T2	1B to 1J		21,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	2					
T3	1C to 1J		21,2,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	3					
T4	1D to 1J		21,2,3,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	22					
T5	1E to 1J		21,2,3,22,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	25					
T6	1F to 1J		21,2,3,22,25,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	5					
T7	1G to 1J		21,2,3,22,25,5,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	24					
T8	1H to 1J		21,2,3,22,25,5,24,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	4					
T9	2A to 2J		21,2,3,22,25,5,24,4,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	27	8				
T10	2B to 2J		21,2,3,22,25,5,24,4,27,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	7					
T11	2C to 2J		21,2,3,22,25,5,24,4,27,7,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	6					
T12	2D to 2J		21,2,3,22,25,5,24,4,27,7,6,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	26					

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TABLE XX. Propagation delay time (t_{PHL}, V_{DD} = 10V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T13	2E to 2J	21,2,3,22,25,5,24,4,27,7,6,26,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	30	8	26	260	ns	14,15
T14	2F to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	29					
T15	2G to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	9					
T16	2H to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	28					
T17	3A to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	11	13				
T18	3B to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,12,32,15,34,14,33,35,16,36,17,20,19,38,18	31					
T19	3C to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,32,15,34,14,33,35,16,36,17,20,19,38,18	12					
T20	3D to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,15,34,14,33,35,16,36,17,20,19,38,18	32					
T21	3E to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,34,14,33,35,16,36,17,20,19,38,18	15					
T22	3F to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,14,33,35,16,36,17,20,19,38,18	34					
T23	3G to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,33,35,16,36,17,20,19,38,18	14					
T24	3H to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,35,16,36,17,20,19,38,18	33					

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TABLE XX. Propagation-delay time (t_{PHL}, V_{DD} = 10V) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T25	4A to 4J	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10V	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 16,36,17,20,19,38,18	35	37	26	260	ns	14,15
T26	4B to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,36,17,20,19,38,18	16					
T27	4C to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,17,20,19,38,18	36					
T28	4D to 4J		21,2,3,22,25,5,24.4,27,7,6,26, 30,29,9,28,11,31,12,32,15,34,14, 33,35,16,36,20,19,38,18	17					
T29	4E to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,19,38,18	20					
T30	4F to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,38,18	19					
T31	4G to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,18	38					
T32	4H to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38	18					

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TABLE XXI. Propagation delay time (t_{PLH}, V_{DD} = 15V) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U1	1A to 1J	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15V	2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	21	23	15	150	ns	14,15
U2	1B to 1J		21,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	2					
U3	1C to 1J		21,2,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	3					
U4	1D to 1J		21,2,3,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	22					
U5	1E to 1J		21,2,3,22,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	25					
U6	1F to 1J		21,2,3,22,25,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	5					
U7	1G to 1J		21,2,3,22,25,5,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	24					
U8	1H to 1J		21,2,3,22,25,5,24,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	4					
U9	2A to 2J		21,2,3,22,25,5,24,4,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	27	8				
U10	2B to 2J		21,2,3,22,25,5,24,4,27,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	7					
U11	2C to 2J		21,2,3,22,25,5,24,4,27,7,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	6					
U12	2D to 2J		21,2,3,22,25,5,24,4,27,7,6,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	26					

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TABLE XXI. Propagation delay time (t_{PLH}, V_{DD} = 15V) test - Continued.

Test condition	Parameter	Connect. A to pin	Connect. B to pin	Connect. C to pin	Initial limits 25°±5°C ambient. Min	Initial limits 25°±5°C ambient. Max	Unit	Test figure
U13	2E to 2J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15V	21,2,3,22,25,5,24,4,27,7,6,26,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	30	8	15	150	ns	14,15
U14	2F to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	29					
U15	2G to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	9					
U16	2H to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	28					
U17	3A to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	11	13				
U18	3B to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,12,32,15,34,14,33,35,16,36,17,20,19,38,18	31					
U19	3C to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,32,15,34,14,33,35,16,36,17,20,19,38,18	12					
U20	3D to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,15,34,14,33,35,16,36,17,20,19,38,18	32					
U21	3E to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,34,14,33,35,16,36,17,20,19,38,18	15					
U22	3F to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,14,33,35,16,36,17,20,19,38,18	34					
U23	3G to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,33,35,16,36,17,20,19,38,18	14					
U24	3H to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,35,16,36,17,20,19,38,18	33					

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TABLE XXI. Propagation delay time (t_{PLH}, V_{DD} = 15V) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U25	4A to 4J	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15V	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 16,36,17,20,19,38,18	35	37	15	150	ns	14,15
U26	4B to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,36,17,20,19,38,18	16					
U27	4C to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,17,20,19,38,18	36					
U28	4D to 4J		21,2,3,22,25,5,24,4,27,7,6,26, 30,29,9,28,11,31,12,32,15,34,14, 33,35,16,36,20,19,38,18	17					
U29	4E to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,19,38,18	20					
U30	4F to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,38,18	19					
U31	4G to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,18	38					
U32	4H to 4J		21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38	18					

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TABLE XXII. Propagation delay time (t_{PHL}, V_{DD} = 15V) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
V1	1A to 1J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	2,3,22,25,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	21	23	20	200	ns	14,15
V2	1B to 1J	21,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	2					
V3	1C to 1J	21,2,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	3					
V4	1D to 1J	21,2,3,25,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	22					
V5	1E to 1J	21,2,3,22,5,24,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	25					
V6	1F to 1J	21,2,3,22,25,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	5					
V7	1G to 1J	21,2,3,22,25,5,4,27,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	24					
V8	1H to 1J	21,2,3,22,25,5,24,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	4					
V9	2A to 2J	21,2,3,22,25,5,24,4,7,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	27	8				
V10	2B to 2J	21,2,3,22,25,5,24,4,27,6,26,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	7					
V11	2C to 2J	21,2,3,22,25,5,24,4,27,7,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38,18	6					
V12	2D to 2J	21,2,3,22,25,5,24,4,27,7,6,30,29, 9,28,11,31,12,32,15,34,14,33,35, 16,36,17,20,19,38,18	26					

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TABLE XXII. Propagation delay time (t_{PHL}, V_{DD} = 15V) test - Continued.

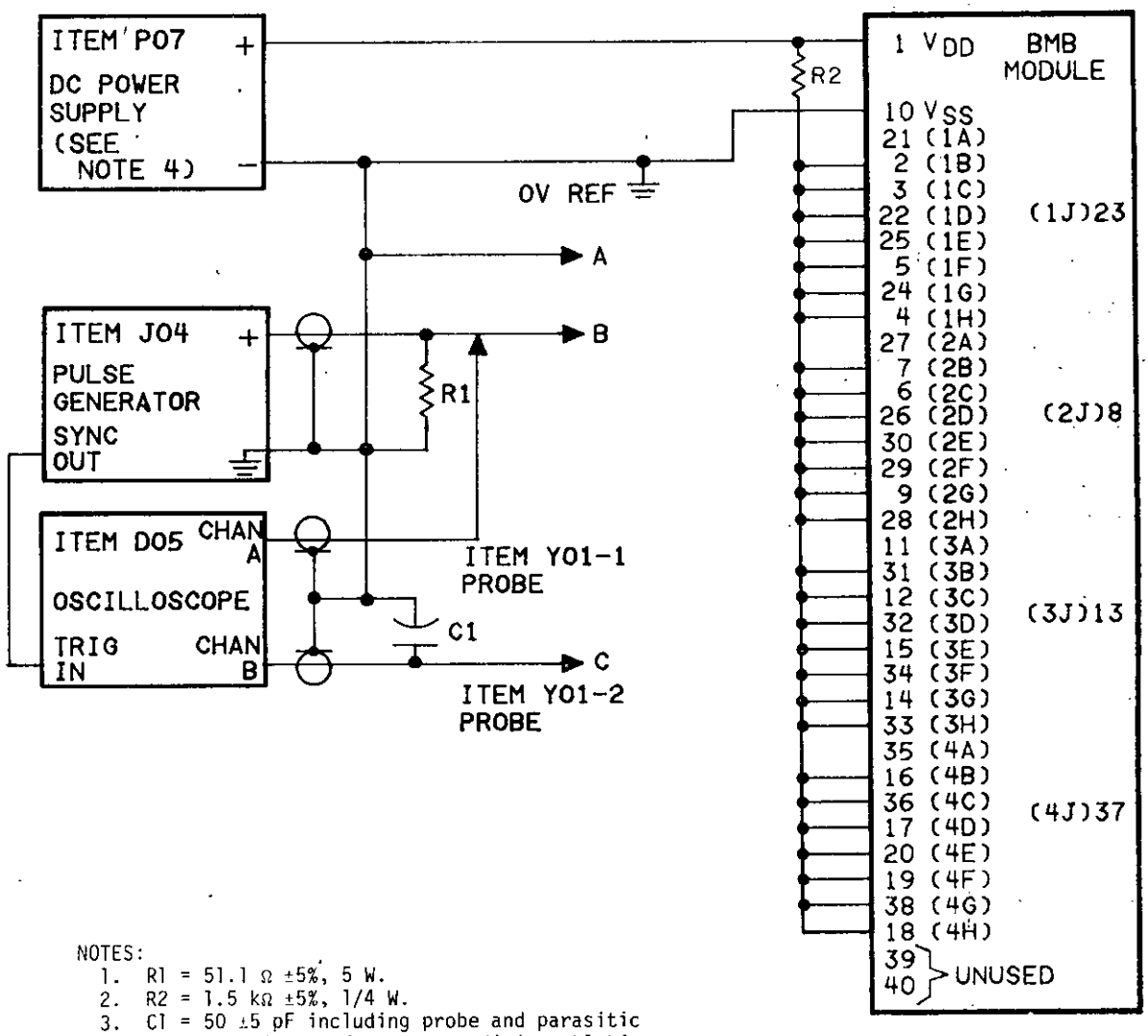
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
V13	2E to 2J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	21,2,3,22,25,5,24,4,27,7,6,26,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	30	8	20	200	ns	14,15
V14	2F to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	29					
V15	2G to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	9					
V16	2H to 2J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	28					
V17	3A to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	11	13				
V18	3B to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,12,32,15,34,14,33,35,16,36,17,20,19,38,18	31					
V19	3C to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,32,15,34,14,33,35,16,36,17,20,19,38,18	12					
V20	3D to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,15,34,14,33,35,16,36,17,20,19,38,18	32					
V21	3E to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,34,14,33,35,16,36,17,20,19,38,18	15					
V22	3F to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,14,33,35,16,36,17,20,19,38,18	34					
V23	3G to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,33,35,16,36,17,20,19,38,18	14					
V24	3H to 3J	21,2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,35,16,36,17,20,19,38,18	33					

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TABLE XXII. Propagation delay time (t_{PHL}, V_{DD} = 15V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V25	4A to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 16,36,17,20,19,38,18	35	37	20	200	ns	14,15
V26	4B to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,36,17,20,19,38,18	16					
V27	4C to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,17,20,19,38,18	36					
V28	4D to 4J	21,2,3,22,25,5,24,4,27,7,6,26, 30,29,9,28,11,31,12,32,15,34,14, 33,35,16,36,20,19,38,18	17					
V29	4E to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,19,38,18	20					
V30	4F to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,38,18	19					
V31	4G to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,18	38					
V32	4H to 4J	21,2,3,22,25,5,24,4,27,7,6,26,30, 29,9,28,11,31,12,32,15,34,14,33, 35,16,36,17,20,19,38	18					

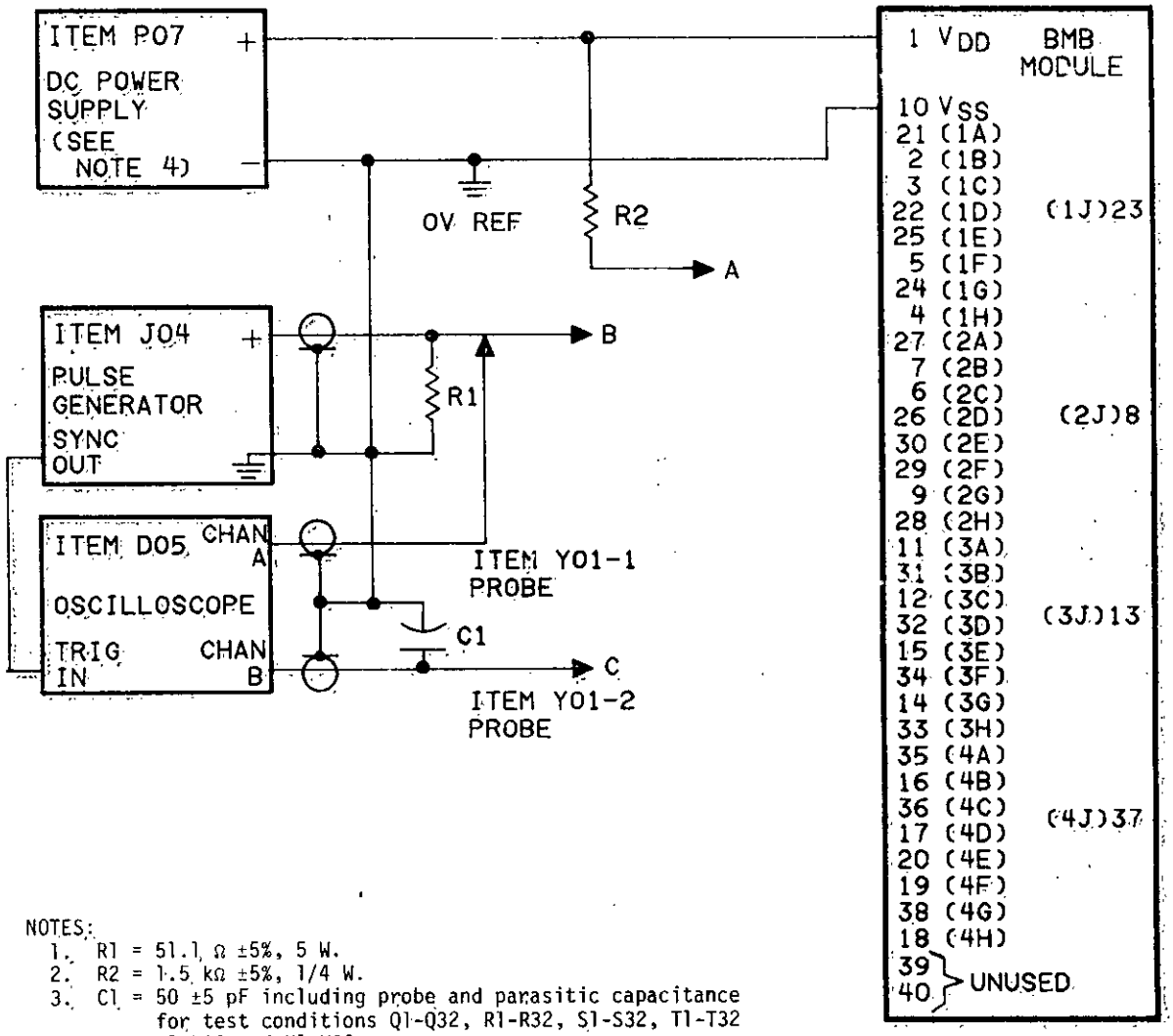
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NOTES:
1. R1 = 51.1 Ω ±5%, 5 W.
2. R2 = 1.5 kΩ ±5%, 1/4 W.
3. C1 = 50 ±5 pF including probe and parasitic capacitance for test conditions L1-L4, M1-M4, N1-N8 and P1-P8.
   = 150 ±15 pF including probe and parasitic capacitance for test conditions L5-L8 and M5-M8.
4. Set item P07 voltage as follows:
   V_{DD} = 5 V ±1% for test conditions L1-L8 and M1-M8.
   = 10 V ±1% for test conditions N1-N8.
   = 15 V ±1% for test conditions P1-P8.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) test circuit for tables XIII through XVI.

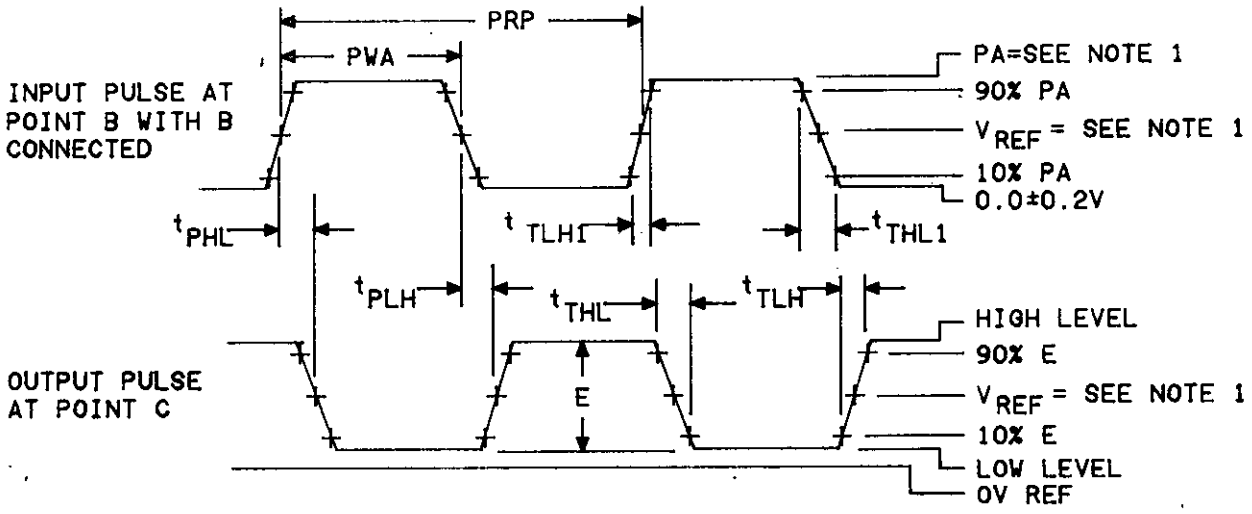
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NOTES:
1. R1 = 51.1 Ω ±5%, 5 W.
2. R2 = 1.5 kΩ ±5%, 1/4 W.
3. C1 = 50 ±5 pF including probe and parasitic capacitance for test conditions Q1-Q32, R1-R32, S1-S32, T1-T32 U1-U32 and V1-V32.
   = 150 ±15 pF including probe and parasitic capacitance for test conditions Q33-Q64 and R33-R64.
4. Set item P07 voltage as follows:
   V_{DD} = 5 V ±1% for test conditions Q1-Q64 and R1-R64.
   = 10 V ±1% for test conditions S1-S32 and T1-T32.
   = 15 V ±1% for test conditions U1-U32 and V1-V32.

FIGURE 14. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVII through XXII.

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NOTES:

1. Input pulse parameters:
   PRP = 5.0 ±0.5 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PWA = 2.5 ±0.2 µs for test conditions L1-L8, M1-M8, Q1-Q64 and R1-R64.
   = 1.0 ±0.1 µs for test conditions N1-N8, P1-P8, S1-S32, T1-T32, U1-U32 and V1-V32.
   PA = 5.0 ±0.1 V for test conditions L1-L8, M1-M8, Q1-Q64 and R1-R64.
   = 10.0 ±0.1 V for test conditions N1-N8, S1-S32 and T1-T32.
   = 15.0 ±0.1 V for test conditions P1-P8, U1-U32 and V1-V32.
   V_{REF} = 2.5 ±0.1 V for test conditions L1-L8, M1-M8, Q1-Q64 and R1-R64.
   = 5.0 ±0.1 V for test conditions N1-N8, S1-S32 and T1-T32.
   = 7.5 ±0.1 V for test conditions P1-P8, U1-U32 and V1-V32.
2. Before making the measurements, verify that the output pulse high-level and output pulse low-level are as follows:

Test conditions	High level	Low level
L1-L8, M1-M8, Q1-Q64, R1-R64	≥ 4.6 V	≤ 0.4 V
N1-N8, S1-S32, T1-T32	≥ 9.5 V	≤ 0.5 V
P1-P8, U1-U32, V1-V32	≥ 13.5 V	≤ 1.5 V

3. The lead length of electrical conductors between termintor resistor R1 and input pins, probes YO1-1 and YO1-2 to input and output pins and between load circuit and output pins shall be less than one inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIII through XXII.

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TABLE XXIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W	Power supply filter capacitance	1	Open	0.007	0.013	µF	16

TABLE XXIV. Input capacitance test.

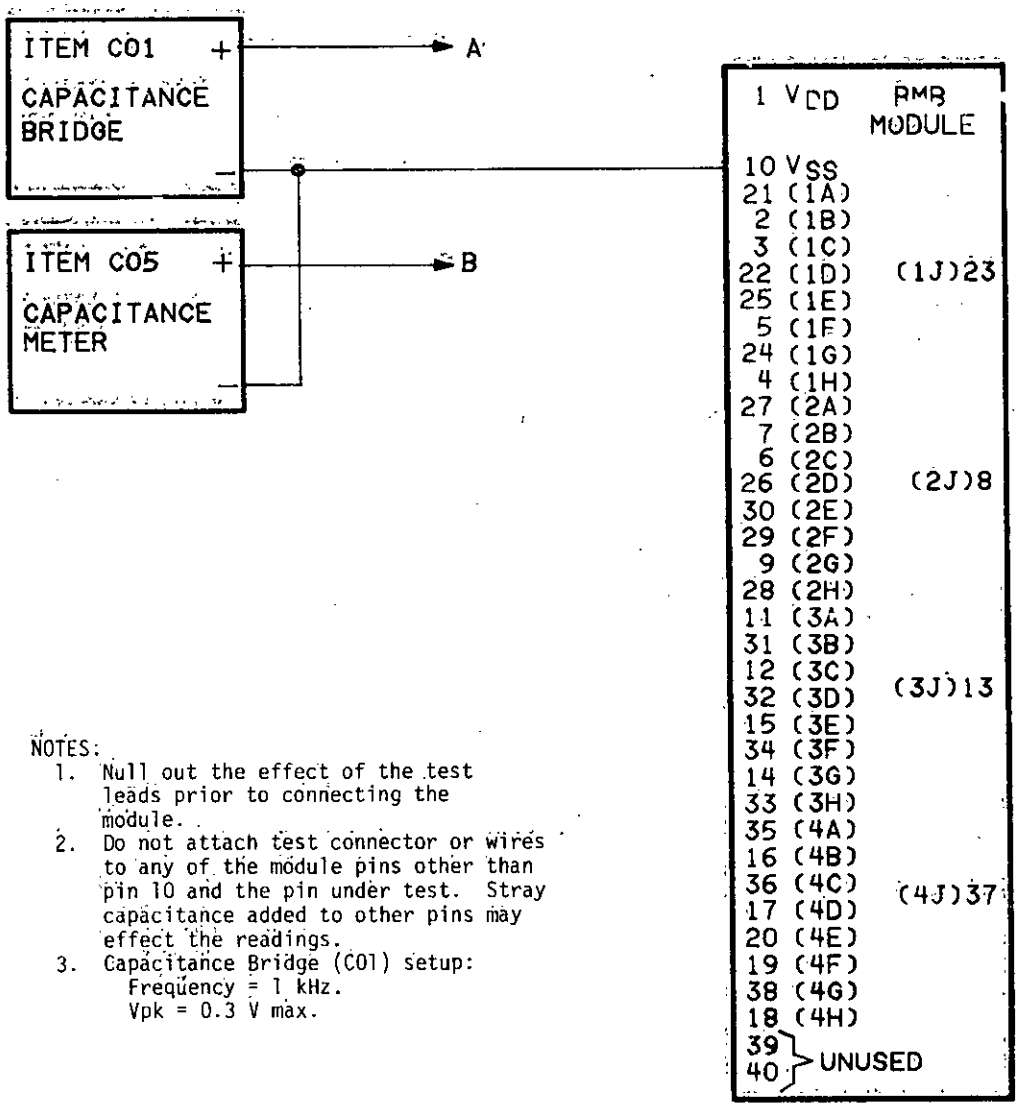
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	1A Input capacitance	Open	21	2	15	pF	16
X2	1B		2				
X3	1C		3				
X4	1D		22				
X5	1E		25				
X6	1F		5				
X7	1G		24				
X8	1H		4				
X9	2A		27				
X10	2B		7				
X11	2C		6				
X12	2D		26				
X13	2E		30				
X14	2F		29				
X15	2G		9				
X16	2H		28				
X17	3A		11				
X18	3B		31				
X19	3C		12				
X20	3D		32				
X21	3E		15				
X22	3F		34				

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TABLE XXIV. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X23	3G Input capacitance	Open	14	2	15	pF	16
X24	3H		33				
X25	4A		35				
X26	4B		16				
X27	4C		36				
X28	4D		17				
X29	4E		20				
X30	4F		19				
X31	4G		38				
X32	4H		18				

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NOTES:
1. Null out the effect of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may effect the readings.
3. Capacitance Bridge (C01) setup:
   Frequency = 1 kHz.
   Vpk = 0.3 V max.

FIGURE 16. Power supply filter and input capacitance test circuit for tables XXIII and XXIV.

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